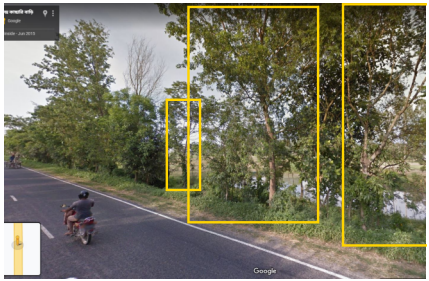
Data

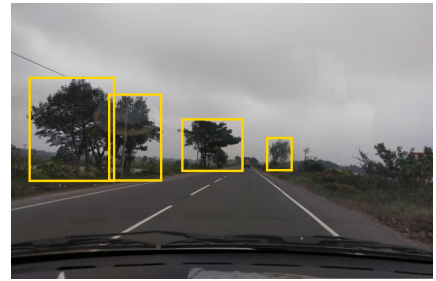
a Bangladesh street photos
benchmark: 1,765 images, 5,311 boxed trees



b Pasadena panoramas
test bed: ○ city-inventory trees projected

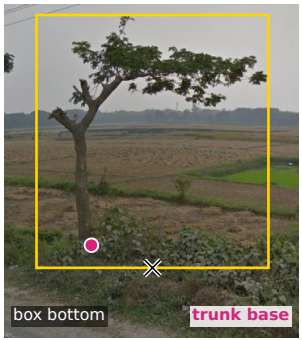


c Dhaka–Chattogram highway
application: Mapillary photos, 240 km



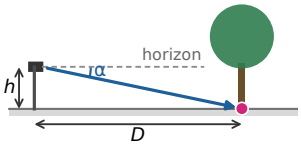
Pipeline (no tree height, no depth network)

d Trunk base
one ground point per tree

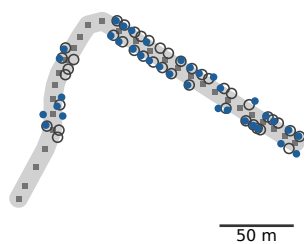


e Range
from the known camera height

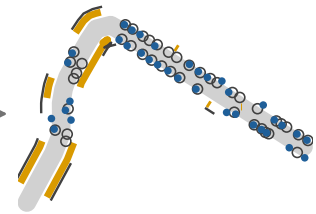
$D = h / \tan \alpha$
camera height known;
tree height not needed



f Merge views into trees
● mapped ○ inventory ■ panorama



g Empty kerb
gold: ours, dark: inventory (S = 12 m)



h Each output is checked against an independent answer key

City tree inventory, Pasadena
map score AP@4 m 0.64
empty length 49 % truly empty
(17 % by chance)

positions, gaps

Field distances, Bangladesh
one phone photo: 2.3 m median error
(1.4 m after scale correction);
phone GPS: 7.0 m

22 trees, ARuler

Marked trees, highway photos
59 % found by the adapted detector
(38 % before adaptation)
→ trees per km

356 trees, 83 photos

Satellite canopy + Landsat heat
tree density vs canopy ρ = 0.48
hottest 10 km: 1.7 vs 22.5 trees/km
in the coolest 10 km

128 km stretches